The book Ramsey number $R(B_4, B_7) = 22$

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Formal proof in Lean 4 with SAT certificates

Abstract. The book B_n is n triangles sharing one edge. We prove $R(B_4,B_7) = 22$. This closes the gap {22, 23} left by Lidický, McKinley, Pfender and Van Overberghe [1], whose flag-algebra bound gave $R(B_4,B_7) \le 23$. The lower bound is their 21-vertex colouring. For the upper bound, a degree-peeling argument reduces the problem to 22 row-fixed SAT instances. The two hardest are split into 312 and 13 cubes, each with a certified coverage proof. All 347 LRAT certificates are checked inside Lean 4 against CNFs that Lean itself generates. A Lean-proved semantic bridge connects these CNFs to a statement file written from the definition. The final theorem `R_B4_B7_eq : LowerBound 21 4 7 ∧ UpperBound 22 4 7` has no `sorry`. It depends only on Lean's standard axioms and the trust in `native_decide`. An independent LRAT checker also accepted all 347 certificates.

1. Introduction

For graphs G and H, R(G,H) is the least N such that every red/blue colouring of the edges of K_N contains a red G or a blue H. Small book Ramsey numbers are tabulated in Radziszowski's survey [2], Table IXa. The value $R(B_4,B_7) \in \{22,23\}$ was listed as open in [1] and in the 2026 revision of [2]. The same pipeline settled $R(B_4,B_9) = 27$ in a companion note [3]. Note that $R(B_4,B_7)$ is not covered by the recent theorem $R(B_{n-2},B_n) \le 4n-3$ of Dai and Lin [4], which addresses books differing by two.

Theorem 1. $R(B_4,B_7) = 22$.

A colouring of K_N is *(m,n)-good* when every red edge has fewer than m common red neighbours and every blue edge has fewer than n common blue neighbours. Thus $R(B_m,B_n) > N$ if and only if K_N has an (m,n)-good colouring. The Lean statement file (SHA-256 `88a1e1b3…d643`) defines this property directly by counting codegrees, and states:

```
def LowerBound N m n := ∃ red, Symm red ∧ BookFree red m n
def UpperBound N m n := ∀ red, Symm red → ¬ BookFree red m n
```

2. Lower bound

LMPV [1] give a (4,7)-good colouring of K_{21}. We embed it as a 21×21 matrix and check it in Lean by evaluation (`lmpv_lower_bound`).

3. Reduction by degree peeling

Let F(d,lo,hi) be the CNF of Section 4. It asserts an (4,7)-good colouring of K_{22} with every red degree in [lo,hi], in which vertex 0 has red degree d, the red neighbours of vertex 0 are 1,…,d, and lex-leader constraints hold inside the blocks {1,…,d} and {d+1,…,21}.

Lemma 2 (row exclusion; Lean `row_excl_of_check`). If F(d,lo,hi) is unsatisfiable, then no good colouring with all red degrees in [lo,hi] has a vertex of red degree d.

The proof relabels the vertex to 0 and its red neighbours to 1,…,d. It then takes a lexicographically maximal representative under the permutations of the two blocks, and that representative satisfies the lex-leader constraints. Lean proves this once for all N, m, n, d, lo, hi. The peeling chain below shrinks the admissible degree interval until it is empty.

Table 1. Peeling chain for N = 22, (m,n) = (4,7).

step	*instances (d; lo, hi)*	*degrees afterwards*	*certificate*
S1	d ∈ {0..5, 14..21}; [0,21]	[6,13]	monolithic
S2	(13;6,13), (12;6,12), (11;6,11)	[6,10]	monolithic
S3	(6;6,10)	[7,10]	monolithic
S4	(10;7,10)	[7,9]	312 cubes + coverage
A1	(7;7,9)	[8,9]	monolithic
A2	(8;8,9)	[9,9]	3 cubes + coverage
A3	(9;9,9)	∅	10 cubes + coverage

We first tried the max-side case (9;7,9). It produced a monolithic proof above 6 GiB, so we peeled from the minimum side instead. A 9-regular graph on 22 vertices is not excluded by parity, so A3 is a genuine case. It is refuted by its certificate.

4. Encoding and cubes

The encoding uses:

- one variable per edge;
- Tseitin variables for red and blue triangles, shared by the three edges of each triple;
- sequential counters bounding the monochromatic codegree of each edge, and the red degree of each vertex;
- unit clauses fixing row 0, and the block lex-leader clauses.

Instances have 45k–59k variables and 101k–127k clauses. On known values the encoding gives the expected answers: (3,5) is SAT at 16 vertices and UNSAT at 17, and (4,6) is SAT at 21 and UNSAT at 22. The same Lean pipeline reproves $R(B_4,B_6) \le 22$ (`R_B4_B6_le_22`).

Cases S4, A2 and A3 use cube-and-conquer. March generated the cubes, and hard cubes were split again. Lean's `unsat_of_cubes` states: if $F \wedge c$ is UNSAT for every cube c, and $F \wedge \wedge_c \neg c$ is UNSAT, then F is UNSAT. The coverage formula is itself refuted by an LRAT certificate checked in Lean.

5. Certificates and verification

Every instance was solved by CaDiCaL with direct binary LRAT output, then trimmed with `lrattrim`. The trimmed text LRAT for S4 alone is about 40 GB. In Lean, certificates are embedded as base64(gzip(LRAT)) and decoded by a pure-Lean inflate. Soundness does not depend on the decoder,

because the only fact used is `LRAT.check cert F = true` for the Lean-generated F. A faulty decoder can only make a check fail. The S4 batches dominate the build: 95 modules, about 18 h at two concurrent builds.

`#print axioms R_B4_B7_eq` lists `propext`, `Classical.choice` and `Quot.sound`, plus one auxiliary axiom per `native_decide` computation. No `sorryAx` appears. Separately, a C LRAT checker sharing no code with CaDiCaL or Lean accepted all 347 trimmed certificates.

During the final build, the compiled Lean outputs for 144 S4 cube proofs were lost after their text proofs had been deleted to save disk. Those 144 cubes were re-solved from the same Lean-generated CNFs. The new proofs again passed the independent checker and Lean. The manifest records both the original and the regenerated hashes.

6. Negative results

The following attempts did not succeed:

- exhaustive circulant and 2-block-circulant searches on 22 vertices, and searches with six prescribed automorphism groups, found no good colouring;
- one-vertex extensions of the LMPV witness all failed;
- Goodman-count cuts gave no speedup;
- monolithic Kissat and CaDiCaL runs on S4 timed out at 6 h;
- unrestricted cube-and-conquer on the full problem (546 cubes) stalled on a hard block.

A complete log of attempts is kept with the artifacts.

7. Scope

This settles one table entry. As with $R(B_4,B_9)$, the methodological point is the end-to-end Lean chain from the raw definition, and its scaling to tens of gigabytes of proof text. We checked arXiv through October 2026 and found no other proof. We cannot exclude unpublished work.

References

[1] B. Lidický, G. McKinley, F. Pfender and S. Van Overberghe, *Electronic Journal of Combinatorics* 32(4) (2025), P4.64.

[2] S. P. Radziszowski, Small Ramsey numbers, *Electronic Journal of Combinatorics*, Dynamic Survey DS1.

[3] Devin (Cognition AI), The book Ramsey number $R(B_4,B_9) = 27$, October 2026.

[4] Dai and Lin, arXiv:2606.07214 (2026).

[5] W. J. Wesley, arXiv:2410.03625 (2024).